6.2 The Natural Base e

Name _____ Period ____ Date _____

Algebra II 6.2 (30-minute plan) · Student sheet

I can…

- define and use the natural base e.
- graph natural base functions and tell growth from decay.
- solve real-life problems with continuously compounded interest.

Vocabulary · fill in the missing word

Natural base e — an _____ number, e ≈ 2.71828, the value $(1 + \frac{1}{n})^n$ approaches as n gets very large

Natural base exponential function — $y = ae^{rx}$: _____ when a > 0 and r > 0, decay when a > 0 and r < 0

Continuously compounded interest — A = _____

Part 1 · Notes and examples (fill in each note, then work the example)

The natural base e: e ≈ _____. It is the value $(1 + 1/n)^n$ approaches as n _____.

Example 1

Simplify each expression.

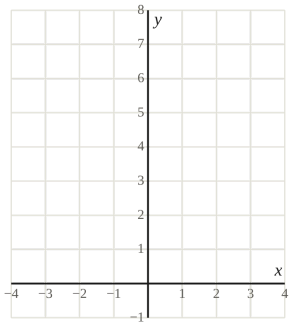
a. $e^3 \cdot e^5$ b. $\frac{16e^5}{4e^4}$ c. $(3e^{-4x})^2$

Growth or decay: $y = ae^{rx}$ (a > 0) shows growth when _____ and decay when _____.

Example 2

Tell whether each function represents exponential growth or decay. Then graph them.

a. $y = 3e^{2x}$ b. $y = e^{\wedge}(-0.5x)$



Rewriting: $e^{rt} = (e^r)^t$, so the growth factor is _____.

Continuously compounded interest: A = _____

Example 3

You deposit $4,000 in an account that pays 5% annual interest compounded continuously. Find the balance after 6 years.

Part 2 · Partner practice

Partner practice

Simplify $(2e^{3x})^3$.